King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 101 Major Exam I Term 243

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 90 minutes

---

1. $\lim_{x\to-\infty} \dfrac{\sqrt{16x^4+2x^3}+5x^2}{3x^2-2} =$

(a) $\dfrac{4}{3}$

(b) 1

(c) 3

(d) $-\dfrac{1}{2}$

(e) $\dfrac{5}{3}$

2. $\lim_{x\to\frac{\pi}{4}} \dfrac{\sin^3 x-\cos^3 x}{\sin x-\cos x} =$

(a) 3

(b) 1

(c) $\dfrac{1}{4}$

(d) $\dfrac{3}{2}$

(e) $\dfrac{1}{2}$

3. The set of all values of x on which the function $f(x)=\dfrac{x}{(x-1)\sqrt{x+1}}$ is continuous is

(a) $(1,\infty)$

(b) $(-\infty,-1)\cup(-1,\infty)$

(c) $[-1,1)$

(d) $(-1,1)\cup(1,\infty)$

(e) $(-1,0)\cup(0,1)\cup(1,\infty)$

4. Which one of the following statements is FALSE? ($\lim_{x\to a} f(x) = \infty$ and $\lim_{x\to a} g(x) = -\infty$, a is a real number)

(a) $\lim_{x\to a}[f(x) - g(x)] = \infty$

(b) $\lim_{x\to a}[-3f(x) + 4g(x)] = -\infty$

(c) $\lim_{x\to a}[g(x)]^2 = \infty$

(d) $\lim_{x\to a} \dfrac{a}{f(x)} = 0$

(e) $\lim_{x\to a}[f(x) + g(x)] = 0$

5. The graph of the function $f(x) = \dfrac{x^2 - 4x + 3}{2x^2 - x - 1}$ has

(a) Two vertical asymptotes and one horizontal asymptote.

(b) one vertical asymptote and no horizontal asymptote.

(c) Two vertical asymptotes and two horizontal asymptotes.

(d) No vertical asymptote and one horizontal asymptote.

(e) One vertical asymptote and one horizontal asymptote.

6. The Intermediate Value Theorem guarantees that the function $f(x) = x^3 - 2x + 3$ has a zero in the interval

(a) $[-1, 0]$

(b) $[-2, 0]$

(c) $[0, 1]$

(d) $[1, 2]$

(e) $[-3, -2]$

7. $\lim_{x\to 0} \dfrac{x \sin x}{1 - \cos x} =$

(a) 0

(b) 1

(c) 2

(d) ∞

(e) $\dfrac{1}{3}$

8. If $\sqrt{x} + \dfrac{1}{x} \leq h(x) \leq x + e^{x-1}$ for all $x > 0$, then $\lim_{x\to 1} h(x) =$

(a) 1

(b) 0

(c) does not exist

(d) 2

(e) 3

9. $\lim_{x\to 3^-} \dfrac{1-2x}{x-3} =$

(a) $-\dfrac{1}{3}$

(b) ∞

(c) 0

(d) $-\infty$

(e) -2

10. $\lim_{x\to 2^-} \dfrac{x^2 - \lfloor x \rfloor}{2x + \lfloor 1 - x \rfloor} =$

(a) $\dfrac{1}{2}$

(b) $\dfrac{2}{5}$

(c) $\dfrac{5}{3}$

(d) 1

(e) $\dfrac{3}{5}$

11. If

$$f(x) = \begin{cases} \dfrac{2x^2 - 9x + 4}{x - 4}, & x \neq 4 \\ 7a, & x = 4 \end{cases}$$

is continuous at $x = 4$, then $a =$

(a) $\dfrac{2}{7}$

(b) 0

(c) $\dfrac{1}{7}$

(d) 2

(e) 1

12. If the line $y = 2x - 4$ is tangent to the graph of $f(x) = kx^3$, then $k =$

(a) $\frac{4}{3}$

(b) $\frac{2}{27}$

(c) $\frac{4}{9}$

(d) $\frac{2}{9}$

(e) $\frac{1}{3}$

13. If $y = Ax + B$ is an equation of the tangent line to the graph of $f(x) = \frac{4}{3x+1}$ at the point $(-1, -2)$, then $AB =$

(a) 6

(b) -12

(c) -3

(d) -9

(e) 15

14. $\lim_{x\to 2} \frac{x-2}{\sqrt{4x+8}-2x} =$

(a) $\frac{8}{3}$

(b) 0

(c) $-\frac{2}{3}$

(d) $\frac{1}{3}$

(e) $-\frac{4}{3}$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(c)**	Divide by x^2: $\frac{\sqrt{16+2/x}+5}{3-2/x^2} \to \frac{9}{3} = 3$.
2	**(d)**	Difference of cubes, cancel: $\sin^2 x + \sin x \cos x + \cos^2 x \to \frac{3}{2}$.
3	**(d)**	Need $x > -1$ and $x \neq 1$: $(-1, 1) \cup (1, \infty)$.
4	**(e)**	$f + g$ is $\infty - \infty$, indeterminate; e.g. $\frac{2}{x^2} - \frac{1}{x^2} \to \infty$, not 0.
5	**(e)**	Factor $(x-1)$ cancels: one VA ($x = -\frac{1}{2}$), one HA ($y = \frac{1}{2}$).
6	**(b)**	$f(-2) = -1 < 0 < 3 = f(0)$; the other intervals show no sign change.
7	**(c)**	Multiply by $1 + \cos x$: $\frac{x}{\sin x}(1 + \cos x) \to 1 \cdot 2 = 2$.
8	**(d)**	Both bounds tend to 2 at $x = 1$; Squeeze Theorem gives 2.
9	**(b)**	Numerator $\to -5$, denominator $\to 0^-$, so the limit is $+\infty$.
10	**(d)**	$\lfloor x \rfloor = 1$, $\lfloor 1 - x \rfloor = -1$ near 2^-: $\frac{4-1}{4-1} = 1$.
11	**(e)**	$\lim_{x \to 4}(2x - 1) = 7 = 7a$, so $a = 1$.
12	**(b)**	$kx_0^3 = 2x_0 - 4$, $3kx_0^2 = 2$ give $x_0 = 3$, $k = \frac{2}{27}$.
13	**(e)**	$A = f'(-1) = -3$, line $y = -3x - 5$, so $B = -5$ and $AB = 15$.
14	**(c)**	Conjugate: denominator $-4(x-2)(x+1)$, limit $\frac{8}{-12} = -\frac{2}{3}$.